SANJUKTA APARNA VENKATACHALAM

Electronics & Embedded Systems Engineer · Firmware · FPGA/RTL · Hardware Bring-Up· DSP · Wireless/IoT

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Right to work: UK Graduate Route to 2028 - no sponsorship required

PERSONAL STATEMENT

Electronics and embedded systems engineer who takes systems from datasheet to working board - bare-metal ARM Cortex-M4F firmware, FreeRTOS, register-level peripheral drivers, a custom UART bootloader, and Verilog RTL on FPGA - validated with instrument-driven debugging on live hardware. Author of a patent-filed LoRa fault-detection network. Top-20-of-800 finalist at the NESIA National Engineering Innovation Challenge. Methodical, validation-first and a fast independent learner.

SKILLS

Embedded & Architecture: ARM Cortex-M4F — exception model, NVIC, MSP/PSP, memory map, MPU; bare-metal register-level programming, FreeRTOS, HAL interoperability, interrupts, timers, PWM, RTC, DMA, low-power modes

Protocols: SPI, I2C, CAN, UART/USART (register + HAL level); LoRa 868 MHz, Wi-Fi

Languages: C, Embedded C, C++, Python, Verilog RTL, ARM Assembly, MATLAB

FPGA/RTL: Verilog RTL, Intel Cyclone V, Quartus Prime, ModelSim

Debug & Tools: Saleae logic analyser, oscilloscope, SWD/JTAG (ST-Link), SEGGER SystemView, ITM/SWV printf, Wireshark; GCC/ARM, custom linker scripts, CMSIS, STM32CubeIDE, Keil uVision, Git/GitHub

Hardware & Practice: Hardware bring-up & schematic reading, sensor integration (IMU/MPU6050) with Kalman filtering, requirements-first validation, technical documentation

KEY ENGINEERING PROJECTS

Broken Power-Line Detection over LoRa - MSc Dissertation ***GitHub***

ESP32 · RFM95W 868 MHz · Raspberry Pi · Python / Flask · MATLAB · Patent filed

- Designed and built a LoRa fault-detection network end to end - two ESP32 nodes relaying events over an 868 MHz link to a Raspberry Pi gateway for segment-level localisation, from MATLAB link-budget modelling to hardware; the link budget supports a range of up to ~32 km.
- Implemented the wireless protocol from scratch - custom packet header, node addressing, private sync word, interrupt-driven receive with a heartbeat / miss-limit scheme (heartbeat every 30 s) for silence-based fault detection.
- Wrote the Python gateway service and Flask dashboard for live node status, event logging and email alerts; validated against requirements before build to avoid rework.

Bare-Metal STM32 Peripheral Driver Library ***GitHub***

Embedded C · Cortex-M4F · register level, no HAL or CubeMX

- Developed register-level drivers for seven STM32 peripherals (GPIO, RCC, SPI, I2C, USART, EXTI, DMA) from the reference manual - blocking, interrupt-driven and slave-mode APIs.
- Verified each driver against live bus traffic on an 8-channel Saleae logic analyser, confirming timing and data integrity before integration; version-controlled throughout with Git.

FlashForge - Custom STM32 Bootloader with UART Firmware Update ***GitHub***

Embedded C · STM32 Cortex-M4F · UART · internal flash programming · Python host tool

- Engineered a custom bootloader across 10 milestones – custom linker scripts, a register-level flash driver, a framed CRC-32 UART protocol, header validation and the bootloader-to-app jump; receives an image over UART, programs internal flash, validates the header and hands control to the app.
- Debugged a handover fault where interrupts left disabled during the jump hung HAL_Delay; fixed by clearing pending NVIC interrupts and re-enabling IRQs before the jump.
- Built a Python host tool and demonstrated failsafe recovery from a deliberately interrupted update.

Hardware-Accelerated Pre-Trade Risk Filter — GitHub

Verilog RTL · Intel Cyclone V · Quartus Prime · ModelSim

- FPGA filter that vets keypad-entered trade orders against 10 hardwired risk rules (quantity, price bounds, halted stock) and returns a PASS/FAIL verdict in ~20 ns; verified in ModelSim before Cyclone V bring-up.
- Built the timing and display path - a hardware timer, a 5-state control FSM, binary-to-BCD conversion and a seven-segment driver presenting the verdict.

Cortex-A9 NEON SIMD DSP Acceleration — GitHub

ARM Cortex-A9 · NEON SIMD intrinsics · C · DE1-SoC

- Bare-metal DSP suite (audio/SDR FIR, FFT, Kalman, image median filter) benchmarking scalar C against NEON vectorised kernels for correctness and speed.
- Vectorised the SDR FIR filter with NEON for a 5x speed-up over scalar, then profiled a median-filter case that ran at 0.75x, pinpointing the cause as branch divergence and non-contiguous memory access.

ADDITIONAL FIRMWARE WORK

FreeRTOS on Cortex-M4 - integrated the kernel into a from-scratch STM32 project (GCC/ARM_CM4F portable layer, heap scheme, linker settings); built multi-task applications with priority pre-emption and ISR-driven priority switching, traced live over UART with SEGGER SystemView.

Real-Time IMU Sensor Fusion - sampled an MPU6050 over I2C on a data-ready interrupt, fused accelerometer/gyroscope data with a Kalman filter under a semaphore/ISR FreeRTOS architecture and drove PWM output from the estimate; tuned live in STM32CubeMonitor.

EDUCATION

MSc Embedded Systems Engineering - University of Leeds, UK — **Sep 2025 – Sep 2026**

Predicted Distinction

Relevant Modules: Embedded Microprocessor System Design, FPGA Design for SoC, Control Systems Design, Wireless Communication System Design, Data Communications and Network Security, Medical Electronics and E-Health

B.E. Electrical & Electronics Engineering - Alagappa Chettiar Govt. College, India — **Nov 2021 – Apr 2025**

CGPA 8.13 / 10

WORK EXPERIENCE

Engineer Tomorrow Scheme - Parker Hannifin, UK — **Feb 2026 – Apr 2026**

Selected onto a competitive industry scheme: designed a conceptual filtration system in CAD, applying engineering design and optimisation principles, and supported peers through technical problem-solving.

Engineering Intern - Neyveli Lignite Corporation (NLC) India Ltd. — **Jun 2024**

Placed on a 210 MW+ thermal generation site: studied turbine control, substation automation and SCADA monitoring alongside senior engineers, analysing real-time fault-protection schemes in live operation.

Embedded / IoT Intern - Verzeo — **Oct 2022 – Nov 2022**

Developed Arduino and ESP32 IoT prototypes with cloud integration for remote monitoring and data visualisation, validating reliable data transmission across each build.

Part-time (University of Leeds conferences, Catering, First Direct Arena, Primark) alongside MSc study.

ACHIEVEMENTS & LEADERSHIP

- **Patent filed** - Smart Electric Pole Fault Detection System (under examination). **Top 20 of 800 teams** at the NESIA National Engineering Innovation Challenge.
- **Outstanding Poster Presentation**, International Leadership Summit, Leeds.
- **1st and 2nd Prize** in consecutive years, College Project Expo (Thiran).
- **Event Head**, EEE Association - three public events, 500+ attendees. **Associate Student Member**, IEEE Women in Engineering. **Volunteer**, National Service Scheme.